Preparing Red Cabbage pH Indicator

Aim: To understand how indicators work.

Equipment (to prepare the indicator)

- Safety goggles
- Red cabbage
- Grater or knife
- Boiled water (kettle)
- Filter paper
- Funnel
- 2 x 500ml glass beaker

Equipment (to use the indicator)

- 5 x 250ml beakers
- Marker pen
- 5 chemicals to test (choose from):
    - Ammonia (NH3)
    - Baking soda
    - Sodium carbonate
    - Lemon juice
    - Vinegar
    - Seltzer water (carbonic acid, H2CO3)
    - Hydrochloric acid

Safety Information



Acids and alkalis - causes irritation to skin, eyes and respiratory organs. Harmful if inhaled or swallowed.

Wear goggles at all times.

Glassware / grater – risk of cuts.

Apply pressure to wounds and inform teacher immediately.

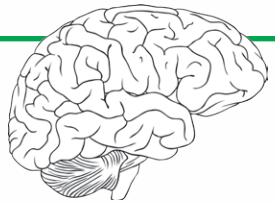
Boiled water – risk of scalds or burns.

Rinse all burns with cold running water for 10 minutes and inform your teacher immediately.

Method

1. Step 1: Grate or chop the red cabbage into small pieces
2. Step 2: Fill the 500ml beaker with the chopped cabbage to about the 400ml mark
3. Step 3: Pour boiled water over the cabbage up to the 500ml mark
4. Step 4: Leave to stand for at least ten minutes until the water has changed colour
5. Step 5: Carefully filter off the chopped vegetables and collect the solution in the other 500ml beaker

visit twinkl.com



How does it work?

Red cabbage contains a pigment molecule which changes colour depending on the pH of a solution.

Very acidic solutions cause the pigment to turn red.

Neutral solutions cause the pigment to turn purple.

Alkali solutions cause the pigment to turn green / yellow.

Questions:

1. Colour the table to complete an indicator scale for the red cabbage indicator.

pH	2	4	6	8	10	12
Colour	red	purple	violet	blue	blue-green	green-yellow

2. Use your indicator scale and choose 6 chemicals to test their pH:
   a. Number the small beakers 1, 2, 3, 4 and 5
   b. Pour 20 – 50ml of the cabbage solution into each of the smaller beakers (how much you add will depend on how much solution you managed to filter from the cabbage mix)
   c. To each beaker add a different solution to find its pH
   d. Record your results in the table

Chemical name	Colour change	pH	Acid, alkali or neutral?

visit twinkl.com

Answers

Questions:

1. Colour the table to complete an indicator scale for the red cabbage indicator.

pH	2	4	6	8	10	12
Colour	red	purple	violet	blue	blue-green	green-yellow

2. Use your indicator scale and choose 6 chemicals to test their pH:
   a. Number the small beakers 1, 2, 3, 4 and 5
   b. Pour 20 – 50ml of the cabbage solution into each of the smaller beakers (how much you add will depend on how much solution you managed to filter from the cabbage mix)
   c. To each beaker add a different solution to find its pH
   d. Record your results in the table

Chemical name	Colour change	pH	Acid, alkali or neutral?
Ammonia	**Green-yellow**	**12**	**Alkali**
Baking soda	**Blue-green**	**10**	**Alkali**
Sodium carbonate	**Green-yellow**	**12**	**Alkali**
Lemon juice	**Red**	**2**	**Acid**
Vinegar	**Purple**	**4**	**Acid**
Seltzer water	**Purple**	**4**	**Acid**
Hydrochloric acid	**Red**	**2**	**Acid**